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Deep and surface learning: a simple or simplistic dichotomy?

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Abstract

In recent years, broad evaluations of higher education in several countries have called for a greater degree of deep learning (i.e. learning with understanding) relative to surface learning (i.e. rote learning). These concepts, having been developed in the 1970s and 1980s, are now well established in the higher education literature. However, to a large extent, exploration and discussion of these concepts is missing from the accounting education literature. The present paper aims to fill this gap in the accounting education literature by introducing the full complexity of this important education literature on deep and surface learning. We show that the use of this dichotomy, which is often used as a convenient shorthand, generally oversimplifies in two key respects. First, the deep–surface distinction is relevant in analysing the following aspects of learning: student learning intentions, learning styles, learning approaches adopted and learning outcomes. The specific context in which the distinction is being applied must be defined carefully. Moreover, it is unrealistic to assume that a deep approach to learning is universally desirable, since it may be necessary, given the nature of the knowledge to be acquired, to adopt a surface approach. Second, the deep–surface approach to learning has been shown to comprise only one of several components which influence a student's overall learning orientation. As a consequence, the identification of several learning orientations, rather than two approaches, gives richer insights into students' learning processes. These orientations, which comprise the learning style and learning approach, are determined partly by the student's personality, motivation and study methods and partly by contextual factors such as the learning task, the attitudes and enthusiasms of the lecturer and the forms of assessment. If intervention strategies designed to improve teaching and learning are to be successful, then a fuller understanding is required of the complex, composite and contingent nature of deep and surface learning and its interrelationship within the teaching-learning environment. This paper calls for a programme of research in accounting education which investigates these issues empirically.

Keywords: accounting education, deep learning, learning approach, learning orientation, learning style, surface learning

Introduction

Significant changes in the business and accounting environment have led to calls for curricular and pedagogical reform in accounting education. Recent changes in the US

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emerged from the American Accounting Association (AAA) Committee on The Future Structure, Content and Scope of Accounting Education (The Bedford Report) (1986). This report called for a flexible and broad-based curriculum which minimized the teaching of procedural accounting, for students to take an active rather than passive role and for a focus on 'learning to learn'. Broadly, the same conclusions were reached by the senior partners of the (then) Big Eight accountancy firms (Arthur Andersen *et al.*, 1989). In response to these documents, in 1989 the AAA appointed the Accounting Education Change Commission (AECC) to create a catalyst for improving the academic preparation of accountants.

Similar developments are occurring in Australia (Mathews, 1990, p. 207) and are supported by the International Federation of Accountants (1994). It is important to recognize, however, that a significant difference exists between the nature of accounting education in the US compared with Australia and also Canada and the UK. In the US, universities rather than the professional institutes are responsible for preparing candidates for professional examinations. As a consequence, university courses in the US have traditionally placed less emphasis on theory and conceptual understanding.

A fundamental common feature of these documents is the belief that there is a need to move away from procedural learning towards a more conceptual form of learning. In order to achieve this, it is argued that it is necessary to assist students to achieve 'deep' rather than 'surface' learning. Deep learning refers to learning with understanding, while surface learning refers to more temporary learning (Williams, 1992, p. 45). These concepts, which were developed and refined during the 1970s and 1980s, are now firmly established in the higher education literature, where the focus is now on ways of promoting and assessing deep learning (Boud, 1990). Despite this, most of the documents which are discussed above do not even mention deep and surface learning explicitly. Williams *et al.* (1988), however, who produced a framework for the development of accounting education research (published by the American Accounting Association), described this as 'a very basic issue which we believe should be discussed' (p. 121).[1]

The general aim of this paper is to introduce this 'very basic issue' into the accounting education literature. We provide a review and synthesis of the theory and empirical evidence from education psychology relating to deep and surface learning.[2] The paper traces the evolution of the concepts of deep and surface learning through the work of four research groups which reveals the concepts' complex and contingent nature and demonstrates that analyses assuming a simple dichotomy will, generally, be inadequate. To be successful, it is likely that proposed strategies for improving accounting education will have to be based on a fuller understanding of the nature of these concepts and their relationship to other elements of the teaching–learning environment.

The main body of the paper identifies the four main research groups which have contributed to the literature on deep and surface learning, outlines the deep–surface distinction and analyses the conceptual and methodological differences between the groups. The analysis reveals five key developmental stages. The final section summarizes the findings and concludes by proposing areas for future research.

[1] A related area of learning theory, learning styles, has, in contrast, been the subject of considerable interest in accounting education. For a recent review of the literature see Wilson and Hill (1994).

[2] Such a review is complicated by the fact that different researchers have applied different terms to essentially the same concepts, (with some using finer classifications of the theoretical constructs) or have identified overlapping concepts. Where possible, we use the terms 'deep' and 'surface' throughout to ensure consistency, with synonymous terms being noted.

The concepts of deep and surface learning

Research into deep and surface learning has focused on student approaches to learning and has been conducted by four main groups of researchers, which are defined with respect to co-authorship networks. During the 1970s research developed to a significant extent independently, however since the 1980s each group has been aware of the work being undertaken elsewhere. The four main groups are as follows.

1. The Lancaster Group, led by Entwistle.
2. The Australian Group, led by Biggs.
3. The Swedish Group, led by Marton.
4. The Richmond Group, led by Pask.

The work of these groups has identified and explored the nature of the fundamental distinction between 'deep' and 'surface' approaches to learning[3].

The deep approach, which implies that a student learns for understanding, is characterized by students who (1) seek to understand the issues and interact critically with the contents of particular teaching materials, (2) relate ideas to previous knowledge and experience and (3) examine the logic of the arguments and relate the evidence presented to the conclusions. The surface approach, which implies that a student learns simply to memorize facts, is characterized by students who (1) try simply to memorize parts of the content of teaching materials and accept the ideas and information given without question, (2) concentrate on memorizing facts without distinguishing any underlying principles or patterns and (3) are influenced by assessment requirements. (Entwistle and Ramsden, 1983; Marton *et al.*, 1984).

Surprisingly, empirical studies in accounting education do not appear to examine or even consider students' approaches to learning. One notable exception is Gow *et al.* (1994) who used Biggs's Study Process Questionnaire to measure students' learning approaches. These learning approches are then related to teaching context variables.

The 'defining features' of approaches to learning have emerged from a fundamental debate in the literature as to whether a student's approach to learning is fixed or variable, i.e. is an inherent individual characteristic or is a response to a situation. As educational psychologists, the natural tendency of Entwistle and Biggs was to explain approaches to learning in terms of consistent traits within individuals, i.e. personality, motivation and attitudes, assuming that other contextual factors, such as the nature of the learning task, would not exert a significant influence. They adopted quantitative methods of investigation based on traditional psychometric techniques such as factor analysis to develop inventories of learning characteristics. It was left initially to other researchers, such as Marton and Pask, to explore the relevance of task perception, task definition, teaching methods and assessment procedures to the learning approach adopted. Typically, these researchers employed qualitative methods, for example Marton utilized a phenomenographical approach.

The Lancaster Group – project 1

The first major series of studies relating to what was later to be regarded as learning approaches was conducted by the Lancaster Group between 1968 and 1973 (Entwistle and Wilson, 1970, 1977; Entwistle *et al.*, 1974). A major objective of these studies was to identify

[3] The terms deep and surface were first used by Craik and Lockhart (1972) with reference to the level of active cognitive processing.

student attributes (i.e. personality, motivational and study method variables) which could be used to predict academic success measured in terms of degree classifications.

An inventory was developed to measure motivation and study methods. The relationship between personality type and study methods was explored by applying cluster analysis to the inventory scores of 1087 first-year undergraduates. Four distinctive student 'types' were identified, each of which was, to some extent, associated with the level of academic achievement. Group 1 was outstandingly successful and motivated by ambition or 'hope for success'; they did not have an active social or sporting life nor had they concentrated on developing aesthetic interests, but they were emotionally stable and had good study methods. Group 2, which was moderately successful, was in many ways the opposite of group 1 as they suffered from self-doubt and 'fear of failure'. However, in spite of low motivation and poor study methods they reached above average levels of attainment by working long hours and adhering very closely to the syllabus requirements. This group had an active social life but only limited aesthetic interests. Group 3, which was also moderately successful, had high motivation and good study methods but did not adhere very closely to the syllabus requirements. This group had high aesthetic values and radical ideals. Finally group 4, the least successful, had active social or sporting interests combined with very low motivation, poor study methods and few hours spent studying (Entwistle and Ramsden, 1983, pp. 33–5).

This research programme did not examine explicitly the processes or strategies used by students in carrying out their everyday academic work; rather it inferred these by relating the students' personality, motivation and study methods to academic performance.

The Australian Group – Biggs

Biggs proposed a model in which the study process mediates between presage factors which exist before the student enters the learning situation and are identified as personal characteristics (i.e. cognitive style,[4] IQ, personality and home background) and institutional characteristics (i.e. subject area, teaching method, assessment method and course structure) and product factors which are identified in terms of academic performance, defined either objectively (for example, examination marks) or subjectively (for example, satisfaction with whatever level of performance is attained) (Biggs, 1978). Biggs (1978) proposed that each of

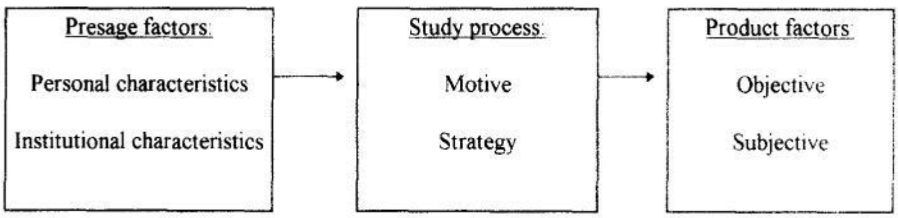

Fig. 1. Biggs's (1978) learning model

[4] In earlier work, Biggs'(1970a,b) hypothesized that a student's study behaviour mediated the connection between cognitive style (seen as the primary student attribute) and performance. Cognitive style is defined as an individual's distinctive way of acquiring, storing, retrieving and transforming information (Kogan, 1973). It is a trait characteristic and is thus not influenced by situational factors. To test his hypothesis Biggs (1970a,b, 1976) devised a Study Behaviour Questionnaire (SBQ) which was administered to students. This instrument contained ten scales which measured personality, cognitive style and study skills.

Table 1. Impact of motive and strategy on approach to learning

Study process		Approach to learning
Motive	Strategy	
Main purpose is to meet minimum requirements: need to achieve balance between working too hard and failing.	Reproductive: limit target to bare essentials and reproduce through rote learning.	Surface (utilizing)
Studies to realize interest and competence in particular academic subjects.	Reads widely with previous relevant knowledge.	Deep (internalizing)
Competitive: tries to obtain highest grades whether or not the material being studied is interesting.	This is the 'model' student: organizes time and working space.	Achieving

Adapted from Biggs (1979).

the presage factors can affect academic performance by affecting students' motives for undertaking learning and by affecting the strategies adopted in approaching learning. The key components of this model, together with their constituent variables, are summarized in Figure 1.

Different combinations of motives and strategies produce different types of learner, all of whom may be equally successful in higher education. The study process is therefore expressed in terms of the motives for learning a task and the strategies employed to realize these intentions, with the related approaches to learning being expressed as motive–strategy combinations, as shown in Table 1. Originally, Biggs (1978, 1979) labelled these approaches utilizing, internalizing and achieving. He later renamed them surface, deep, and achieving, respectively, to achieve greater consistency with other writers' terms (Biggs, 1987). Effective learning was expected to require congruence between the motive and strategy, i.e. matching; this is Biggs's congruence hypothesis.

Deep and surface approaches describe the ways in which students engage in the context of the specific task to be accomplished, whereas the achieving approach describes the way in which students organize their time and working environment. It is also possible that students adopt mixed approaches to learning. For example, an individual engaged in rote learning in a highly organized way is adopting a surface-achieving strategy, reading for meaning in an organized way is deep-achieving, while rote learning and seeking meaning (an unlikely simultaneous combination) is surface–deep. Biggs developed the Study Process Questionnaire (SPQ) to test this model on higher education students. Empirical studies by Biggs using this inventory provide satisfactory evidence on the reliability, internal consistency and construct validity of its constituent scales (Biggs, 1982; Biggs and Kirby, 1983). SPQ scale scores are related to educational, demographic and individual difference variables. High 'deep approach' scores, for example, were obtained by arts faculty students, students in their first year, students who planned further formal education and students who were older.

Biggs (1987) later developed an elaborated model of student learning based on the research of his and other groups. This model uses the concept of metacognition, which is defined as an individual's awareness of his or her cognitive resources. Biggs (1987) proposed the term metalearning for the application of metacognition to the specific area of student learning. He provided evidence that, subject to the constraints of their cognitive ability, students are able to choose deliberately the approaches to learning which are most likely to bring about the desired outcome, i.e. students behave metacognitively (Biggs, 1987, Ch. 6). The metalearning ability required by the congruence hypothesis discussed above thus requires an awareness of the available options and the ability to exert control over these options. Biggs (1987) argued that, as the range and power of student ability increases, more options in the learning process are identified. Students discriminate between those options according to their pattern of processing abilities. Metalearning increases as the general cognitive ability of students increases. The other important factor is the locus of control, which describes the extent to which students believe themselves to have some control over their learning (an internal type) compared to those who believe their learning to be governed by external forces (an external type). Biggs (1987) presented evidence which supports the view that approaches to learning vary according to students' capability for metalearning which, in turn, is related to individual student differences such as general ability and the locus of control. 'Deep' and 'achieving' approaches can interact with ability and the locus of control to affect examination performances in ways which suggest that metalearning is present. These interactions may explain partly why the relationship between approaches to learning and performance in some students was found to be very weak, since a specific approach may be effective for some students and ineffective for others.

The Swedish Group

The concepts of deep and surface approaches to learning are also found in the work of the Swedish Group based in Gothenburg. The main series of studies (Marton, 1975; Säljö, 1975; Marton and Säljö, 1976a,b; Svensson, 1976, 1977; Fransson, 1977) examined the learning approaches to and the learning outcomes from reading academic articles. This was seen as one of the main types of learning demanded of students. The instructions about reading the article were deliberately vague. Interviews with 30 students elicited their intention, approach, understanding of the task and what they remembered and understood (Marton and Säljö, 1976a,b). From the analysis of interview transcripts, response patterns were identified and explanatory constructs hypothesized. Regularities in the students' approach to learning and the learning outcome were uncovered. There was a fundamental distinction between deep and surface approaches to learning, with a deep approach resulting in a high, i.e. deep, level of understanding and a surface approach resulting in a low, i.e. surface, level of understanding.[5] This association was argued to be partly inevitable, since the adoption of a deep approach to learning is a necessary but not sufficient condition for a deep understanding (Svensson, 1977).[6]

The Swedish Group's principal contribution was to emphasize that the intention of the student is crucial, in that what a student intends to get out of learning determines whether a deep or surface approach will be used. The approach used, in turn, determines the 'level of performance'. It can be noted that this distinction between intention and approach is paralleled by Biggs's (1987, p. 14) distinction between motive and strategy. This early Gothenburg work stresses that individuals select an approach to studying in response to both a particular task content and a particular context. That is, an individual's study approach is flexible. This contrasts with the view originally proposed by the Lancaster Group, which is

[5] Marton (1975) referred initially to deep and surface levels of 'processing'.

[6] In an independent analysis of transcripts, Svensson (1977) distinguished holistic and atomistic categories, which represented the way in which students organized their descriptions of what they remembered and which overlap broadly with the deep versus surface categorization of approaches to learning.

that individuals have fixed, predetermined study approaches.

A limitation of the early work of the Swedish Group was, however, that it was conducted in artificial circumstances, not in a real studying situation. Later Gothenburg studies addressed this limitation. For instance, Svensson (1977) detected deep and surface study approaches in normal study contexts, finding that students adopting a deep approach tend to spend longer studying, as they find it more interesting.

The Gothenburg researchers went on to consider the way in which the approach to learning adopted was influenced by various contextual factors, in particular the type of question asked and the student's level of interest and anxiety. Säljö (1975) and Marton and Säljö (1976b) found a relationship between learning approach and the type of question given in tests, with a surface approach being easier to induce than a deep approach. The overburdening of syllabuses and examinations with factual information may be responsible for the low levels of understanding exhibited by students when prevented from reproducing answers by well-rehearsed methods. Fransson (1977) found that levels of interest and anxiety can also affect the approach to learning adopted by students. Students who perceive a situation to be threatening, whether intended or not, were more likely to adopt a surface approach. A lack of either interest or perceived relevance also tended to lead to surface learning. Säljö (1979) investigated students' conceptions of learning via interviews and found that sophisticated learners realized that the optimal learning strategy or approach is influenced by context, whereas unsophisticated learners see learning as rote memorization.

The Richmond Group – Pask

Outside these groups, Pask and his associates, working from Richmond, investigated how students learn complex new fields of knowledge, i.e. how students master unfamiliar ideas in an article or textbook (Pask and Scott, 1972; Pask, 1976a,b, 1977; Robertson, 1977). In these experiments, students were, in contrast to the Gothenburg studies, explicitly required to reach a deep level of understanding. Two general categories of learning approaches were identified, termed learning strategies: serialist and holist. The serialist strategy is characterized by a step-by-step approach to reading a text; in adopting this strategy a student will work systematically through the text, perhaps taking copious notes, progressing from one simple hypothesis to another. The holist strategy is characterized by a more complex way of

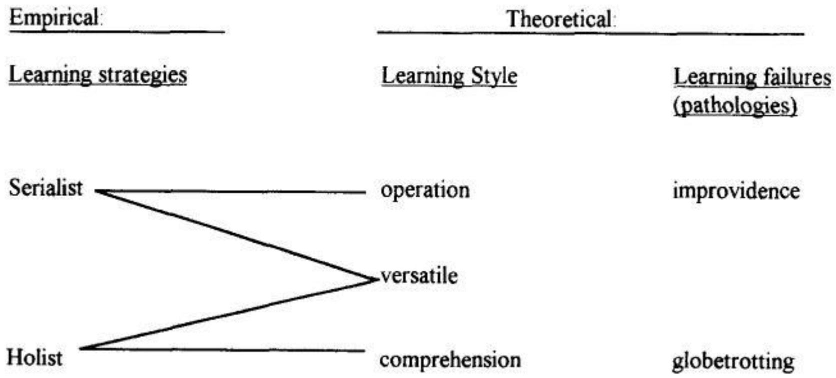

Fig. 2. The Richmond Group's empirical and theoretical learning categories

achieving understanding; in adopting this strategy a student will read around whatever comes to their attention, searching forward and backwards in the text for points of interest, moving from topic to topic in an attempt to obtain the gist of the subject without the chore of reading every word (Pask and Scott, 1972).[7]

Pask (1976b) subsequently suggested that holism and serialism were 'extreme manifestations of more fundamental processes' (p. 133). The general tendency to adopt a particular strategy is referred to by Pask (1977) as a learning style. The holist will adopt a style termed comprehension learning which involves building descriptions of what is known, while the serialist will adopt a style termed operation learning which is concerned with the mastering of procedural details. Students who are able to adopt successfully either style, depending on the subject matter, are termed versatile (Pask, 1977). These differences in learning style are apparent not only in the way understanding is reached, but also in the way it is not reached. Such learning failures are termed pathologies. The first of these, globetrotting, is described by Pask (1976a) as the holist's tendency to make inappropriate or vacuous analogies and/or to generalize from insufficient evidence. The second, improvidence, is the serialist's tendency to fail to use valid analogies and not to build an overall framework linking key elements of the topic. Figure 2 provides a summary of the empirical and theoretical learning categories identified by the Richmond Group and of the links between them.

The Lancaster Group – project 2

The concepts developed by the Swedish and Richmond Groups formed the main theoretical basis for the second 5-year project undertaken by the Lancaster Group in 1975–1980. The group noted that the Swedish Group's deep and surface approaches to learning overlapped significantly with the Richmond Group's holist and serialist learning strategies and comprehension and operation learning styles (Entwistle and Ramsden, 1983, p. 28). The project aimed to extend the work of these two groups both conceptually and empirically using a mixture of quantitative and qualitative methods. Thus, the Lancaster Group now acknowledged the potential influence of external factors on the study approach adopted by students and no longer viewed the study approach as an inherent student characteristic, fixed and determined by individual traits.

The specific objectives of the project were to measure approaches to and styles of studying using an inventory and relate this to (1) cognitive styles, cognitive skills (i.e. level of ability given a particular cognitive style) and personality characteristics, (2) student perceptions of the department's academic 'climate', (3) the task and (4) the academic discipline. This was done using an 'Approaches to Studying Inventory' (ASI). Three successive pilot versions of the ASI were developed before a final version emerged. The second pilot inventory comprised 15 subscales intended to measure motivation, in addition to variables identified by the Australian, Swedish and Richmond Groups and related the results to academic achievement. A factor analysis of 767 first-year undergraduate student responses identified four factors which explained 56% of the total variance in responses. These factors were labelled deep approach/comprehension learning, surface approach/operation learning, organized and achieving and stable extraversion (i.e. sociable students who have no fear of

[7] Pask and Scott (1972) developed a 'conversation theory of learning' which describes how a student works towards a full understanding of a topic by questioning or trying out ideas on a 'third' person. Full understanding occurs only when the student can explain a topic by reconstructing it and can demonstrate this understanding by applying the principles learned to a new situation.

Table 2. Study orientations and characteristic elements

Orientation	Characteristic elements
Meaning	Deep approach, comprehension learning, interrelating ideas, use of evidence and intrinsic motivation
Reproducing	Surface approach, operation learning, improvidence, fear of failure, syllabus boundness and extrinsic motivation
Achieving	Achievement motivation, intrinsic motivation and strategic approach
Non-academic	Disorganized study methods, negative attitudes to studying, globetrotting and low intrinsic motivation

Adapted from Entwistle and Ramsden (1983, Table 4.4 (p. 49), Table 4.5 (p. 52, p. 53).

failure) (Entwistle *et al.*, 1979). The close correspondence between the first three factors and Biggs's (1978, 1979) internalizing, utilizing and achieving factors was noted by the Lancaster Group (Entwistle and Ramsden, 1983, p. 40). In the third pilot inventory, deep–surface learning was related to three separate elements: intention, process and outcome. In addition, the comprehension and operation learning styles were distinguished from their corresponding pathologies (globetrotting and improvidence). Thus, the ideas of the Swedish and Richmond Groups were incorporated explicitly into this version of the inventory. In the final version, two essential elements of a deep level outcome were added to the inventory: 'relating ideas' and 'use of evidence'.

This final inventory was completed by 2208 second-year undergraduates across six disciplines. When the results were assessed in conjunction with those from other studies which had employed the same inventory, it was concluded that four distinctive orientations to studying emerged consistently. Study orientation became a term used to denote the combination of learning style and learning approach, which are themselves influenced by personality, study methods, motivation and strategy. The characteristic elements of each orientation are shown in Table 2, which has been adapted from Entwistle and Ramsden (1983, Ch. 4).

'Meaning orientation' comprises a deep approach, comprehension learning, the ability to relate ideas and use evidence and intrinsic motivation. 'Reproducing orientation' comprises a surface approach, operation learning, improvidence, fear of failure, syllabus boundness and extrinsic motivation. 'Achieving orientation' comprises achievement motivation, intrinsic motivation and a strategic approach. Finally, 'non-academic orientation' comprises disorganized study methods, negative attitudes to studying, globetrotting and low intrinsic motivation. Since the intention to adopt a particular approach to learning reflects, in part, the individual's characteristic learning style, the connection between a deep approach and comprehension learning and between a surface approach and operation learning becomes inevitable (Entwistle and Ramsden, 1983, Ch. 4).

Since the completion of the Lancaster Group's second project, the main focus of the higher education literature has been mainly on ways to promote and assess deep learning (Boud, 1990). Research of a more fundamental nature has been very limited.[8]

[8] Entwistle (Entwistle and Entwistle, 1991; Entwistle, 1995) and Marton (Marton et al., 1993) have focused on the relationship between students' learning orientations and students' understanding and, more broadly, on the relationship between student learning and teaching methods, including forms of assessment. Schmeck *et al.* (1991) explored the interaction between personality (in particular, the student's self-concept) and learning approaches.

Summary and conclusions

This paper has demonstrated that the distinction between deep and surface learning is complex. The complexity of these concepts and their contingent nature became apparent as the development of this line of research was traced from a simple input–output model of student learning (Lancaster Group – project 1), through a model which focused on the learning process (Biggs), through one which incorporated intention (motivation) and contextual factors (Swedish Group), through one which distinguished a student's persistent tendency, i.e. learning style, from the specific approach adopted (Pask), to, finally, a widely applicable general model (Lancaster Group – project 2) in which metalearning acts as the dynamic link between student, task and outcome (Biggs).

Thus this literature, viewed as a whole, demonstrates that a student's approach to learning is only partly a function of his or her general characteristics, since it can be modified by specific learning situations. Such situational influences include the student's perception of the relevance of the learning task, the attitudes and enthusiasm of the lecturer and the expected forms of assessment. The extent to which a student's predilection for a particular approach can be modified is determined by their metalearning capability.

The design of intervention strategies which improve teaching and learning in accounting education will require a sound understanding of the complex and contingent nature of learning approaches. We call for a programme of research in accounting education which investigates empirically the role of learning approaches within the accounting discipline. We identify four general issues to be addressed.

1. It is widely believed that accounting attracts a relatively high proportion of reproducing and achieving students. Does, therefore, the inherent approach to learning of accounting undergraduates differ from that of the general undergraduate population?
2. Since deep understanding involves the appropriate use of both comprehension learning and operation learning, to what extent can the direct teaching of study strategies and specific study skills assist students to become more aware of their own style, to recognize its strengths and weaknesses and to be able to consciously use alternative approaches?
3. Given the inherent mix of procedural and conceptual knowledge in accounting, which teaching methods promote a deep approach to learning? In addition, to what extent is deep learning increased as a result of strategies which integrate the teaching of accounting techniques with the teaching of economics, finance, organization theory, etc., rather than teaching separate functional courses in these subjects?
4. Which forms of assessment (for example, case studies and essays) reward critical thinking and thus encourage a deep approach to learning?

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